Jihong Min

Assistant Professor, Department of Biomedical Engineering
National University of Singapore
N1 Institute for Health, 28 Medical Dr, Singapore 117456
Email: jhmin@nus.edu.sg
Google Scholar

Education

2024 - 2026 iCMB NIH T32 Postdoctoral Research Fellow in Medical and Electrical Engineering
California Institute of Technology
Prof. Azita Emami, Prof. Wei Gao

2018 - 2023 Ph.D. in Medical Engineering
California Institute of Technology
Prof. Wei Gao

2014 - 2017 B.S. in Electrical Engineering
University of Illinois at Urbana-Champaign
Prof. Joseph Lyding

Research & Work Experience

2026-Present — Assistant Professor, Department of Biomedical Engineering, National University of Singapore, Singapore (Presidential Young Professorship)

- Leading independent research group developing ingestible and wearable bioelectronic platforms for multiplexed GI monitoring, gut-brain axis decoding, and closed-loop therapy.
- Building capabilities in ultra-low-power electronics, aptamer/enzyme sensing, and soft robotic capsule actuation, with clinical collaborations at NUHS.

2024-2026 — iCMB NIH T32 Postdoctoral Research Fellow, Caltech, USA (Advisors: Prof. Azita Emami, Prof. Wei Gao)

- Development of ingestible electronic systems with electrochemical sensors based on aptamers and enzymes for real time monitoring of metabolites and neurotransmitters in the gastrointestinal tract.
- Development of a wearable Electrical Impedance Tomography (EIT) belt for accessible liver screening, targeting early detection of non-alcoholic fatty liver disease (NAFLD) with a flexible and fully autonomous system incorporating a conductive polymer hydrogel and a micro-electrode array.

2018-2023 — Graduate Researcher, Caltech, USA (Advisor: Prof. Wei Gao)

- Development of robust and cost efficient wearable triboelectric nanogenerator and a flexible solar cell based on a quasi-2D perovskite layer with α-methylbenzylamine (MBA) as an additive for efficient and sustainable powering of wearables.
- Development of novel electrochemical sensors based on enzymes, ion-selective electrodes, antibodies tagged with gold nanoparticles, and molecularly imprinted polymers on laser printed graphene (LEG) and inkjet printed carbon/gold electrodes for sensitive and selective molecular detection in biofluids.
- Design of miniaturized and low-power wearable/ingestible electronic systems and integration of energy harvesting modules and electrochemical sensors for wireless monitoring of physiochemical biomarkers.

2016-2017 — Undergraduate Researcher, UIUC, USA (Advisor: Prof. Joseph Lyding)

- Self-assembly of centimeter scale polystyrene microsphere monolayers using Langmuir Blodgett transfer.
- Fabrication of gold-graphene-gold bowtie structure for exploration of large scale plasmonic field enhancements.

2016-2016 — Summer Undergraduate Researcher, GIST, South Korea (Advisor: Prof. Jae-Hyung Jang)

- Fabrication of the Metal-Insulator-Metal structure ReRAM devices.

Honors & Awards

- 02/2026. NUS Presidential Young Professorship
- 09/2024. Nature and Health Conference Best Poster Award
- 08/2024. NextProf Nexus Fellow
- 03/2024. iCMB NIH T32 Fellowship, Caltech
- 03/2024. SPIE Soft Mechatronics and Wearable Systems Best Paper Award
- 11/2023. MRS Graduate Student Award
- 09/2018. Andrew and Peggy Cherng Endowment Fellowship, Caltech
- 12/2017. Graduated with High Honors, UIUC

Teaching

Instructor:

- Spring 2027 (Semester 2, AY2026/27). Biomedical Instrumentation. BN3405, NUS

Teaching Assistant:

- Spring 2018. Theory and Fabrication of Integrated Circuits. ECE 444, UIUC

Guest Lecturer:

- Spring 2020. New Frontiers in Medical Technologies. MedE 205, Caltech
- Winter 2021. Sensors in Medicine. MedE 202, Caltech
- Winters 2023, 2024. Sensors in Medicine. MedE 202, Caltech
- Summer 2024. Crash Course on Wearables for DIVE Veteran Students, Caltech

Conference Presentations, Poster Sessions, Invited Talks, and Workshops

- Oral Presentation: SPIE Optics + Photonics, Organic and Hybrid Sensors, Detectors, and Bioelectronics XIX Conference, San Diego, CA, Aug 2026.
- **(Invited)** Seminar: University of California, Los Angeles (UCLA), Bioengineering, Los Angeles, CA, Aug 2026.
- **(Invited)** Seminar: University of California, Irvine (UCI), Mechanical and Aerospace Engineering, Irvine, CA, Aug 2026.
- Oral Presentation: MRS Fall Meeting & Exhibit, Boston, MA, Nov 2025.
- **(Invited)** Oral & Poster Presentation: Nature and Health Conference, Brown University, Providence, RI, September 2024 **(Best Poster Award)**.
- Workshop: NextProf Nexus, University of Michigan, Ann Arbor, MI, August 2024.
- **(Invited)** Seminar: Caltech's DIVE (Diversification Initiative through Veteran Education) program, Pasadena, CA, August 2024.
- **(Invited)** Seminar: Korea Advanced Institute of Science and Technology (KAIST), Chemical Engineering, Daejeon, South Korea, July 2024.
- **(Invited)** Seminar: Gwangju Institute of Science and Technology (GIST), Mechanical Engineering, Gwangju, South Korea, July 2024.
- Oral Presentation: SPIE Soft Mechatronics and Wearable Systems Conference, Long Beach, CA, Mar 2024 **(Best Paper Award)**.
- Poster Presentation: Caltech Medical Engineering 10th Year Anniversary Symposium, Pasadena, CA, May 2024.
- Oral Presentation: MRS Fall Meeting & Exhibit, Boston, MA, Nov 2023 **(Graduate Student Award)**.
- Poster Presentation: IEEE International Flexible Electronics Technology Conference, San Jose, CA, Aug 2023.

Academic Service

Reviewer for Journals and Conferences:

Science Robotics, Nature Communications, Nature Sensors, Biosensors and Bioelectronics, ACS Photonics, ACS Applied Nano Materials, npj Flexible Electronics, npj Flexible Materials, npj Biomedical

Innovations, Electroanalysis, Scientific Reports, Analytical Chemistry, Talanta, IEEE Consumer Electronics Magazine, Electrochemistry Communications, ACS Applied Materials & Interfaces, IEEE Open Journal of Engineering in Medicine and Biology, Sensors & Diagnostics, Electroanalysis.

- Volunteer (Lab Tour), Southern California Science Olympiad, 2025.
- Judge, Orange County Science and Engineering Fair (OCSEF), 2025.
- Judge, Summer Undergraduate Research Fellowships (SURF) Perpall Speaking Competition, Caltech, 2024.
- Instructor, Crash Course on Wearable and Ingestible Sensors for DIVE Veteran Students, Caltech, 2024.

Proposal Writing Experience

Principal Investigator (PI):

- NUS PYP startup grant (Funded)
- ICMB NIH T32 Training Grant proposal (Funded) - T32EB023858
- Merkin Institute for Translational Research Spark Grant proposal

Grant Writing Assistance:

- NIH NIDCD Research Project Grant (R01) proposal (Funded) - R01DC021461
- Office of Naval Research (ONR) proposal (Funded) - N00014-21-1-2483

Patents

- "Systems and Methods for Powering Autonomous Sweat Sensor," W. Gao, J. Min, 2023, US Patent Application No. 18/077,846.
- "Wearable Autonomous Biomimetic Sweat Sensor for Precision Nutrition," W. Gao, M. Wang, Y. Yang, J. Min, 2022, US Patent Application No. 17/824,798.

Selected Publications

(34 papers with 9 as first/co-first author, 10,348 citations, h-index 25, updated 10/2026) † indicates equal contributions

1. **Min, J.**†, Ahn, H.†, Lukas, H., Ma, X., Bhansali, R., Sunwoo, S.-H., Wang, C., Xu, Y., Yao, R., Kim, G., Li, Z., Hsiai, T. K., Emami, A., Jung, H.-T., & Gao, W. (2025). Continuous biochemical profiling of the gastrointestinal tract using an integrated smart capsule. ***Nature Electronics***, 8, 844-855. **Featured on Journal Cover**
2. **Min, J.**†, Demchyshyn, S.†, Sempionatto, J. R., Song, Y., Hailegnaw, B., Xu, C., Yang, Y., Solomon, S., Putz, C., Lehner, L., Schwarz, J. F., Schwarzinger, C., Scharber, M., Shirzaei Sani, E., Kaltenbrunner, M., & Gao, W. (2023). An autonomous wearable biosensor powered by a perovskite solar cell. ***Nature Electronics***, 6, 630-641. **Featured on Journal Cover**
3. Wang, M.†, Yang, Y.†, **Min, J.**†, Song, Y., Tu, J., Mukasa, D., Ye, C., Xu, C., Heflin, N., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2022). A wearable electrochemical biosensor for the monitoring of metabolites and nutrients. ***Nature Biomedical Engineering***, 6, 1225-1235. **Featured on Journal Cover**
4. Song, Y.†, **Min, J.**†, Yu, Y., Wang, H., Yang, Y., Zhang, H., & Gao, W. (2020). Wireless battery-free wearable sweat sensor powered by human motion. ***Science Advances***, 6(40), eaay9842.
5. **Min, J.**†, Tu, J.†, Xu, C.†, Lukas, H.†, Shin, S., Yang, Y., Solomon, S. A., Mukasa, D., & Gao, W. (2023). Skin-interfaced wearable sweat sensors for precision medicine. ***Chemical Reviews***, 123, 5049-5138. **Featured on Journal Cover**
6. Heng, W., Yin, S., **Min, J.**, Wang, C., Han, H., Shirzaei Sani, E., Li, J., Song, Y., Rossiter, H. B., & Gao, W. (2024). A smart mask for exhaled breath condensate harvesting and analysis. ***Science***, 385, 954-961.
7. Tu, J., **Min, J.**, Song, Y., Xu, C., Li, J., Moore, J., Hanson, J., Hu, E., Parimon, T., Wang, T.-Y., Davoodi, E., Chou, T.-F., Chen, P., Hsu, J. J., Rossiter, H. B., & Gao, W. (2023). A wireless patch for the monitoring of C-reactive protein in sweat. ***Nature Biomedical Engineering***, 7, 1293-1306.

8. **Min, J.**, Song, Y., & Gao, W. (2022). Microcracked conductors for wearable sensors. ***Nature Electronics***, 5(11), 717-718.
9. **Min, J.**†, Sempionatto, J. R.†, Teymourian, H.†, Wang, J., & Gao, W. (2021). Wearable electrochemical biosensors in North America. ***Biosensors and Bioelectronics***, 172, 112750.
10. **Min, J.**, Yang, Y., Wu, Z., & Gao, W. (2020). Robotics in the gut. ***Advanced Therapeutics***, 3(4), 1900125.

Other Publications

1. Kim, I., Yao, D. R., Yin, S., Tran-Ngoc, P. T., **Min, J.**, Ng, N., Heng, W., Li, J., Sadri, B., Sato, H., Kim, S. O., & Gao, W. (2026). BeetleBot: An integrated bioinspired soft robot for multimodal sensing and adaptive interaction. ***Science Advances***, 12(34), eaeg8849.
2. Xu, C., Zheng, H., Heng, W., Liu, R., MarionSims, J., Li, J., Jin, P., Tay, R. Y., **Min, J.**, Wang, G., Yue, Y., & Gao, W. (2026). Human-inspired time-series health evaluation with an adaptive multimodal electronic skin. ***Science Advances***, 12(31), eaeg5606.
3. Kim, G., Cho, S., Ahn, H., Wang, C., Han, H., Ma, X., Kim, M.-J., Qu, J., **Min, J.**, & Gao, W. (2026). Strain-isolated microneedles for ambulatory hormonal and metabolic monitoring. ***Nature Nanotechnology***.
4. Gao, B., Ge, Z., Sun, X., Lin, B., Gao, Y., **Min, J.**, Zhang, Q., Rogers, J. A., Yu, X., Gao, W., & Lim, C. T. (2026). AI-powered closed-loop wearable bioelectronics for personalized and autonomous healthcare. ***Nature Sensors***, 1, 667-680.
5. Li, J., Fan, K., Ma, X., Wang, C., **Min, J.**, Chen, Y., Song, Y., Xu, Y., & Gao, W. (2026). Strain-insensitive wet-tissue-adhesive biphasic bioelectronics for physicochemical monitoring and adaptive therapy. ***Nature Materials***.
6. Xu, Y., Ma, X., Fan, K., **Min, J.**, Qu, J., Heng, W., Li, J., Tang, S., Kim, G., Liu, R., Yin, S., Li, W., & Gao, W. (2026). Strain-resilient intrinsically stretchable electrochemical biointerfaces. ***Science***, 392(6801), eaed1630.
7. Heng, W., Putz, C., Tang, W., **Min, J.**, Lehner, L., Demchyshyn, S., Kim, G., Wang, C., Hailegnaw, B., Rippatha, E., Scharber, M., Kim, M., Chen, Y., Fan, Z., Kaltenbrunner, M., Gao, W. (2026). A battery-free smart mask for long-term exhaled breath biochemical sensing. ***Nature Sensors***, 1, 328-340.
8. Shin, S., Liu, R., Yang, Y., Lasalde-Ramírez, J. A., Kim, G., Won, C., **Min, J.**, Wang, C., Fan, K., Han, H., Uwakwe, C., Heng, W., Hsiai, T. K., Li, Z., FitzGerald, J. D., & Gao, W. (2025). A bioinspired microfluidic wearable sensor for multiday sweat sampling, transport, and metabolic analysis. ***Science Advances***, 11(33), eadw9024.
9. Wang, C., Fan, K., Shirzaei Sani, E., Lasalde-Ramírez, J. A., Heng, W., **Min, J.**, Solomon, S. A., Wang, M., Li, J., Han, H., Kim, G., Shin, S., Seder, A., Shih, C.-D., Armstrong, D. G., & Gao, W. (2025). A microfluidic wearable device for wound exudate management and continuous in situ analysis of wound biomarkers. ***Science Translational Medicine***, 17(795), eadt0882.
10. Tu, J., Yeom, J., Chaj Ulloa, J., Solomon, S. A., **Min, J.**, Heng, W., Kim, G., Dao, J., Vemu, R., Pang, M., Wang, C., Kim, D.-H., & Gao, W. (2025). Stressomic: A wearable microfluidic biosensor for dynamic profiling of multiple stress hormones in sweat. ***Science Advances***, 11(32), eadx6491.
11. Wang, M., Ye, C., Yang, Y., Mukasa, D., Wang, C., Xu, C., **Min, J.**, Solomon, S. A., Tu, J., Shen, G., Tang, S., McCune, J. S., Hsiai, T. K., Li, Z., & Gao, W. (2024). Printable molecule-selective core-shell nanoparticles for wearable and implantable sensing. ***Nature Materials***, 24, 589-598.
12. Xu, C., Song, Y., Sempionatto, J. R., Solomon, S. A., Yu, Y., Nyein, H. Y. Y., Tay, R. Y., Li, J., Heng, W., **Min, J.**, Lao, A., Hsiai, T. K., Sumner, J. A., & Gao, W. (2024). A physicochemical-sensing electronic skin for stress response monitoring. ***Nature Electronics***, 7, 168-179.
13. Ye, C., Wang, M., **Min, J.**, Tay, R. Y., Lukas, H., Sempionatto, J. R., Li, J., Xu, C., & Gao, W. (2024). A wearable aptamer nanobiosensor for non-invasive female hormone monitoring. ***Nature Nanotechnology***, 19, 330-337.
14. Song, Y., Tay, R. Y., Li, J., Xu, C., **Min, J.**, Shirzaei Sani, E., Kim, G., Heng, W., Kim, I., & Gao, W. (2023). 3D-printed epifluidic electronic skin for machine learning-powered multimodal health surveillance. ***Science Advances***, 9(37), eadi6492.

15. **Min, J.**, Gao, W. (2023). Battery-Free Wearable Electrochemical Sweat Sensors. ***IEEE IFETC***.
16. Choi, Y., Ho, D. H., Kim, S., Choi, Y. J., Roe, D. G., Kwak, I. C., **Min, J.**, Han, H., Gao, W., & Cho, J. H. (2023). Physically defined long-term and short-term synapses for the development of reconfigurable analog-type operators capable of performing health care tasks. ***Science Advances***, 9(27), eadg5946.
17. Shirzaei Sani, E., Xu, C., Wang, C., Song, Y., **Min, J.**, Tu, J., Solomon, S. A., Li, J., Banks, J. L., & Armstrong, D. G. (2023). A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds. ***Science Advances***, 9(12), eadf7388.
18. Mukasa, D., Wang, M., **Min, J.**, Yang, Y., Solomon, S. A., Han, H., Ye, C., & Gao, W. (2023). A Computationally Assisted Approach for Designing Wearable Biosensors toward Non-invasive Personalized Molecular Analysis. ***Advanced Materials***, 35(35), 2212161.
19. Yu, Y., Li, J., Solomon, S. A., **Min, J.**, Tu, J., Guo, W., Xu, C., Song, Y., & Gao, W. (2022). All-printed soft human-machine interface for robotic physicochemical sensing. ***Science Robotics***, 7(67), eabn0495. **Featured on Journal Cover**
20. Torrente-Rodríguez, R. M., Lukas, H., Tu, J., **Min, J.**, Yang, Y., Xu, C., Rossiter, H. B., & Gao, W. (2020). SARS-CoV-2 RapidPlex: a graphene-based multiplexed telemedicine platform for rapid and low-cost COVID-19 diagnosis and monitoring. ***Matter***, 3(6), 1981-1998. **Featured on Journal Cover**
21. Torrente-Rodríguez, R. M., Tu, J., Yang, Y., **Min, J.**, Wang, M., Song, Y., Yu, Y., Xu, C., Ye, C., & IsHak, W. W. (2020). Investigation of cortisol dynamics in human sweat using a graphene-based wireless mHealth system. ***Matter***, 2(4), 921-937.
22. Yang, Y., Song, Y., Bo, X., **Min, J.**, Pak, O. S., Zhu, L., Wang, M., Tu, J., Kogan, A., & Zhang, H. (2020). A laser-engraved wearable sensor for sensitive detection of uric acid and tyrosine in sweat. ***Nature Biotechnology***, 38(2), 217-224.
23. Yu, Y., Nassar, J., Xu, C., **Min, J.**, Yang, Y., Dai, A., Doshi, R., Huang, A., Song, Y., & Gehlhar, R. (2020). Biofuel-powered soft electronic skin with multiplexed and wireless sensing for human-machine interfaces. ***Science Robotics***, 5(41), eaaz7946.
24. Song, Y., **Min, J.**, & Gao, W. (2019). Wearable and implantable electronics: moving toward precision therapy. ***ACS Nano***, 13(11), 12280-12286.